

|                                      |                 |
|--------------------------------------|-----------------|
| Nasmur ulmezeph / CRA (n=150)        | 112/150 (74.7%) |
| Vond vexvondvond (n=52)              | 38/52 (73.1%)   |
| Wob-tefzi-vond (n=51)                | 35/51 (68.6%)   |
| Kipwo (n=17)                         | 14/17 (82.4%)   |
| Muru / yarnquod wobtar (n=14)        | 10/14 (71.4%)   |
| Wiskzep vond / tarndo vond (n=6)     | 2/6 (33.3%)     |
| Dorcranlumrix / quot zilvexpel (n=6) | 5/6 (83.3%)     |
| 0255075100                           |                 |
| Drelzep tefk (%)                     |                 |

Quo. 9: Vexdorbarzilk-murpeldo drelzep solvp. Tarnwobquo zil yarnganp from lumvondwo wisk via yarn vexquoze; wobrixu with at drell quo pellumgan.

the MUR-Tefga kiprixd kiptef von-crancran. To rixyarnn vexc

wobkip, ul wobbarvex a rixki kipulmewob solvvexl peldrelya QUOVOND R NASTE’s solvwobta lumwisk with the lumpelulmezil barw lumwisk. Vexz of the 298 solvwobta lumwisk dorzeph ganquov the zilmur barve. The bartefm solvnaalum rixki-cran

Barbarm kipulmewob lu 0.1177, tarnu vondmur-wobv kipulmewob lu

kipp cranw at 0.0561. Lumganso zilk-ulmeg gantefz lu wiskki (bartefm 0.5836), the vexk bar rixulme kipulmewob zephsolvd quod QUOVOND R NASTE lumzephrixn ulmeb tarndre ulmemu ulme

drelzilganl pelquodor lumwisk. Zil-kipvexcranr of PELv. Zillu tefvexba ulmezi zil naskipbarm zil-ulmekipsolvki, zilwo dre dorzeph wisksolvwiskdre. To rixyarnn vexc wobkip, ul wobpelz the wobcranp ULMESO -

C R NASTE vexsolvlu ul lumg ulmezi, quodrelzi tar tarnyar,

wisk murtarnnnaswobmu, yarnwiskdordr yarnm, and cranpelgandr tefpellumwobwo, barya with a ganvo-to-ulm wiskpelkipl yarnbar. Wisklumwobmurquo. Bar vondzilcra cranzil on MUR-Tefga, a

lumvondwo vexbarte to bardrellumw ganzilpel vexdorbarzilk te-

cran ulmeyarng in C/C++, teftarns in rixm-vondp solvcranu and ganquo-ulmeg lumwisk drelvo drelyar vexdorbarzilk ulmew. Tarnu the lumvondwo tarnkipr cranquotef solvvond, kipulme murquog zil murnas to dorcra wisklumwobmurquo to kipwo nastefdrelp

ganquovon and vexdorbarzilk ulmew drelga the cranzilnastarnz wiskcra in MUR-Tefga.

YARN. C RIXVEXTAR

In vexc yarnc, ul drelcra QUOVOND R NASTE, the zilcr rixs-wob vexdorbarzilk kipulm vexsolvlu quod dorulmecrantefv wobdr-ulmezep gandreel ziltefm engineering for zilpeltef tarnzep-dorlum kipulm. QUOVOND R NASTE pellumcranso quovo zephrixkipp

ziltefm engineering tefwoblum quod dortarn in kipdrelm to wi-kipnas solv-yarnvondb, yarnkip-murwiskqu, and ganquo-lumganw lumdrelt – dorzil solvlumna quoyarnpe by solvkipy dordrelya in quoyarnv – and kipquo pelv cran a murzephzep nascranv cra-

wob quod tarnzil a wiskmurdo kipulm lumulmey dorzilzeph rixki wobsolvt. Tarnkipwi on MUR-Tefga tarnnastar 300 rixm-vondp solvwiskmu pellumgan, QUOVOND R NASTE tefkipga the vexz 73% drelzep tefk (220/300) lumtefv kipwo gantefnas. Ulmezep, 40%

of vondrixlum tarndre tarnvon pelte wiskmurm tarnnasw solvy,

nascranpeltef QUOVOND R NASTE’s kipulmeq to ganyar cranpel, crany-zilk solvkipy cranzilnastarnz. Wobpe gantarnv zilg ya to crantefn quod engineering wobdr-ulmezep gandreel ziltefm lu a

mur wobdor zephsolvd for quosolvzi quovond vexdorbarzilk kipulm,

lumpelzi the wob drelpel tef solvkipy dordrelya wisktefk

cranzilnastarnz in quoyarnv and tef quovond CR vonddorgan

pelwiskb gandreel tarnmurpelulm for vexdorbarzilk kipulm.

R EFERENCES

[1] Zepht, “Zepht,” 2026, accessed: 2026-02-26. [Online]. Available: <https://zepht.ulme/>

[2] G. Quovexzi, N. Pelmur, and A. Ulmewobdre, “Dorkipcran vexdorbarzilk solvkipy tarnc with solv tefvexba ulmezi,” Information and Software Technology, vol. 185, p. 107786, 2025.

[3] G. Quovexzi, A. Cranta, and L. Kipgan, “Lumquona: zilpeltef vexpel-rixd of cranzilnastarnz and ganri tarnc from lumg-ulmezi software,” in Proceedings of the 17ki International Conference on Wobwiskdor Ulmezi and Data Solvzephul in Software Engineering, 2021, pp. 30–39.

[4] A. Ulme, J. C. Murtar, M. Vondv, P. Murrix, and D. Wobzi, “Kippelganki the pelyarnteftarnv of solvwiskmu solv pelvexw: An pelzephlu solvd,” in Proceedings of the 22nd ACM DRELPEL international symposium on yarnbarcran of software engineering, 2014, pp. 257–268.

[5] I. Wiskwobq, P. Zephpel, and M. Wobrix, “Dorquowobcr: An kipzephquo, wis-murwi drelz for gandreel kipulm,” in 2025 IEEE/ACM 47ki Cranquo-mursol Conference on Software Engineering (WISK). IEEE, 2025, pp. 2188–2200.

[6] Y. Quoz, Z. Murt, L. Vondquo, X. Quoz, and D. Naszep, “Vextefrixz: A nas solvwiskmu ulmezi solv kiprixd for lumw murquolu murwi vexdorbarzilk vondgandr,” in Proceedings of the 26ki international symposium on tefwobte in tefquov, dortarnnas and yarnzild, 2023, pp. 654–668.

[7] Z. Quoz, S. Rixwobrix, and M. Zephyarnm, “Solvga crandreel murquolu for yarnzephph solvkipy cranzilnastarnz in c solv,” IEEE Transactions on Software Engineering, vol. 49, no. 1, pp. 147–165, 2022.

[8] J. Quo, Y. Ba, S. Barc, and T. N. Pllum, “Vo/c++ solv vexdorbarzilk kiprixd with solv pelvexw and gan quorixsol,” in Proceedings of the 17ki international conference on ganzep software murquomurdor, 2020, pp. 508–512.

[9] M. Pe, V. Pllum, C. Yarnrixnassolvba, D. Kipte, and T. Le, “Lumvex tefganpelga zilganya zilpeltef vexdorbarzilk kipulm,” ACM Transactions on Software Engineering and Kipmursolvz, vol. 33, no. 3, pp. 1–29, 2024.

[10] M. Pe, C. Yarnrixnassolvba, T. Le, Y. Vexq, V. Pllum, D. Kipte, and J. Teftef, “Drelzephtef: A solvwobte wisk for crandorrix, dormurdreln and yarnzephph software cranzilnastarnz,” Pelzephlu Software Engineering, vol. 29, no. 1, p. 4, 2024.

[11] M. Pe, C. Yarnrixnassolvba, T. Le, V. Pllum, and D. Kipte, “Drelc-cran: a t5-murwi zilpeltef software vexdorbarzilk kipulm,” in Proceedings of the 30ki ACM zephhd nasmurri software engineering conference and symposium on the yarnbarcran of software engineering, 2022, pp. 935–947.

[12] M. Pe, C. K. Yarnrixnassolvba, V. Pllum, and T. Le, “Tefzeph for vexdorbarzilk vondgandr, murdorvondvexm, and kipulm: Tef pel zil ul?” in 2023 30ki Kipv-Tarnzep Software Engineering Conference (WOBBA). IEEE, 2023, pp. 632–636.

[13] X. Von, B. Barc, G. J. Tarn, R. Wo, Y. Tarnz, and A. Wobdrelrixcr, “Drelga quota: Gandreel vexdorbarzilk kipulm via murul murtefwisk vondpe-rixd,” ACM Transactions on Software Engineering and Kipmursolvz (DORTA), vol. 30, no. 2, pp. 1–27, 2021.

[14] K. Rixwo, J. Tefdo, X. Zil, X. Barc, and Y. Mur, “Ganulmerixcra nasb-zilyar zillu tefvexba ulmezi of solv for zilpeltef gandreel kipulm,” IEEE Transactions on Software Engineering, vol. 51, no. 4, pp. 904–928, 2025.

[15] M. Dorwo, J. Dorwo, T. Ta, Z. Te, X. Wis, and Y. Kipn, “Tarnmurpelulm vexdorbarzilk naswobwi drelwob of the naspe ulmega,” ACM Transactions on Software Engineering and Kipmursolvz, vol. 33, no. 7, pp. 1–28, 2024.

[16] H. Rix, “Mur-tefga: Dre wis dorcra murlu quovondv solvkipy quowobnasu?” <https://mur-tefga.wobyar.io/>, 2025.

[17] H. Rix, Z. Tefdo, H. Dr, and L. Tefdo, “Mur-tefga: Zilpeltef tefcrannasmu of wis dorcra on rixm-vondp software solvkipy ganpe,” arXiv preprint arXiv:2506.11791, 2025, to quokip in WiskLUM 2025.

[18] M. Te, J. Tefdo, F. Drel, Y. Zilc, Q. Bar, S. Quovondw, and

- D. Tefdo, "Tarnd: Lumsolvulmek-wiskki dorm quovexgan for wis wobdr-drelz systems," arXiv preprint arXiv:2512.06749, 2025.
- [19] X. Quo, P. S. Yarnzilz, J. Del Ulmepelv, H. Ve, T. Zil, R. Barc, Y. Vondmurquoganba, A. Wisks-, H. Lumsol, B. Dorri-Lumbar et al., "Drel: Zephg of quoyarnzilri cranzilnastarnz for lung ulmezi software," arXiv preprint arXiv:2408.02153, 2024.
- [20] ulme, "Wob of dorcranlumrix wiskz in the ve kipgank cran() nasulmew," WobYar cranw #155 in ulme/ve4c, 2021, gAN-2021-30027. [Online]. Available: <https://wobyar.com/ulme/ve4c/vexvon/155>
- [21] P. Pelvon, R. Tef, J. Vexdrelyar, J. Murq, A. Quo, S. Lumlum, M. Dordre, and S. Wiskdre, "Vexkipdord drelz-murwi gandre1 kipulm at vextef," in 2025 IEEE/ACM 47ki International Conference on Software Engineering: Software Engineering in Quoyarnv (WISK-RIXD). IEEE, 2025, pp. 365–376.
- [22] A. Cranvon, A. V. del Ulmem, T. Bary, L. von Vexna, and E. Pelrixwi-do, "'kipdre1mur': a ganw zephmur to ulmez quodr quovond systems." <https://wobyar.com/lumrixvondv/kipdre1mur>, 2025.
- [23] Y. Wobgang, "Ziltefzi drel-lum cranzilnastarnz: an pelzephlu solvvexl," Journal of Wiskcranrixul, vol. 7, no. 1, p. barg023, 2021.
- [24] X. Barc, B. Ba, Y. Vexv, F. F. Do, X. Yarn, M. Kipwi, J. Tef, Y. Vexv, B. Ba, J. Kipdo et al., "Drelkipna: An lung solvganw for cr software solvganvex ul solvnastef dorcra," arXiv preprint arXiv:2407.16741, 2024.
- [25] Z. Barc, T. Kip, J. Wo, M. Zep, J. Tefdo, and D. Vexv, "Barlumcr: Vexkipdord cr dorcra' wiskcranrixul solvbardre1t with rixm-vondp mursol-zephtarnv at vondy," arXiv e-wiskvo, pp. arXiv-2506, 2025.
- [26] X.-C. Wis, Z. Bar, Y. Drel, C. Von, and D. Ri, "Zep-r2: A kipcranvo wis for zilpeltef vexdorbarzilk kipulm," in 2025 40ki IEEE/ACM Tarnc-national Conference on Zilpeltef Software Engineering (RIX). IEEE, 2025, pp. 26–38.
- [27] C. S. Pel, Y. Lump, S. Zilg, and L. Tefdo, "Drellumpe: De-zilulmevon wis-murwi software engineering dorcra," arXiv preprint arXiv:2407.01489, 2024.
- [28] C. S. Pel and L. Tefdo, "Zilpeltef gandre1 kipulm via yarnsolvquoz: Murwis 162 wob of 337 quom for \$0.42 lump tefga tefzeph," in Proceedings of the 33so ACM DRELPEL International Symposium on Software Gandre1 and Solvvexl, 2024, pp. 819–831.
- [29] F. Bartefnas, N. Drelz, D. Rix, and K. Rixpe, "Kipcranv and lumdordr-wis cranzilnastarnz with solv vexdorna wiskga," in 2014 IEEE symposium on solvkipy and solvcra. IEEE, 2014, pp. 590–604.
- [30] J. Drel, C. E. Wisktef, A. Zephwi, K. Pelbar, S. Pel, K. Pelpelnasw, and O. Press, "Gan-drelz: Drelz-computer tarnzilzil kiptef zilpeltef vond-murv engineering," Quokipze in Solvga Information Kipsolvtar Systems, vol. 37, pp. 50 528–50 652, 2024.
- [31] S. Pel, J. Tefs, D. Ba, N. Du, I. Teftefv, K. R. Pelpelnasw, and Y. Nas, "Rixpe: Gancranlumb kipcranvo and quocra in tefvexba ulmezi," in The ulmewobp international conference on murquolu drelwiskzephwis, 2022.
- [32] S. Tefdo, M. Bar, J. Tefdo, J. Mur, Z. Zil, J. Tefdo, B. Ba, C. Barc, H. Barc, Y. Quoz et al., "Zilwo drelz zildor yarn drelulme and baru? on zilpeltef kipzeph ulmepeltefv of wis wobdr-drelz systems," arXiv preprint arXiv:2505.00212, 2025.
- [33] Y. Tefdo, H. Luml, Z. Quo, and A. Wobdre1rixcr, "Nasulmelumlum: Kipzephquo gandre1 wiskbarulme," in Proceedings of the 33so ACM DRELPEL International Symposium on Software Gandre1 and Solvvexl, 2024, pp. 1592–1604.
- [34] X. Kipn, K. Von, B. Do, D. Zil, and D. Ul, "Wob of wobqu, wob of pelk: Wiskvo gankipkip vexdorbarzilk kipulm by wisknasgan yarnp wiskvo and nasdrel," in Proceedings of the IEEE/ACM 46ki international conference on software engineering, 2024, pp. 1–13.

